SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Embedded Systems
COURSE CODE: ECA1411

COURSE OUTCOME COVERED

CO1: *Analyze and design complex and networked embedded systems using formal design methodologies, embedded architectures,CAN (BL5)*

Assessment Tool Weightage	: 30%
Assessment Tool	: Concept Mapping
Duration	: 45 minutes
Marks	: 20
Type of Assessment conduction	: Inside Class
Mode of submission	: Hand-drawn on A3/A4 chart using color pens and sketches OR AI-designed color printout
Topics Covered	: Introduction of Embedded Systems, Complex Systems and Microprocessor, Embedded System Design Process, Core components, Formalism for System Design, Embedded product development cycle, Design
Evaluation Type	: Individual Concept Mapping Exercise
Result Generation	: Rubric-Based Evaluation

Description:

- The Concept Mapping Assessment is conducted after the completion of 75% of the syllabus as a Higher-Order Thinking Internal Assessment Tool.
- Each student is assigned one concept mapping question from a set of 20 questions.
- Students visually represent the flow and interconnection of concepts using hand-drawn charts (A3/A4) with colors, sketches, and diagrams, or by creating the map using AI tools and submitting a color printout.
- Each concept map carries 20 marks.
- Evaluation is based on concept accuracy, linkage of ideas, visual clarity, completeness, and creativity.

Total Marks: 20

Sl. No	QUESTION BANK	BLOOMS LEVEL
1	Develop a comprehensive concept map that connects the evolution of embedded systems with advancements in automation, connectivity, and intelligent system behavior, establishing relationships among IoT integration,	BTL 5

Linkages	25 %	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20 %	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20 %	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10 %	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Criteria	Concept Identification	Linkages	Organization	Integration of Literature	Clarity	Total Marks
Weightage (%)	25%	25%	20%	20%	10%	20
Max Mark	5	5	4	4	2	20
Q1	5	5	4	4	4	18
					Total(100):	90

Faculty In Charge
(Dr.PANDI MAHARAJAN M)

EMBEDDED PRODUCT DEVELOPMENT CYCLE

1. CONCEPT DEFINITION
- Product idea
- Purpose
- Feasibility

2. REQUIREMENT ANALYSIS
- User requirements
- Functional needs
- Constraints

3. SYSTEM DESIGN
- Architecture
- Hardware selection
- Software structure

4. HARDWARE - SOFTWARE CO-DESIGN
- HW/SW partitioning
- Interface design
- Integration planning

5. PROTOTYPING
- Build prototype
- Initial integration
- Demonstrate concept

6. TESTING
- Functional testing
- Performance testing
- Find defects

7. VALIDATION
- Verify requirements
- Safety & reliability
- Compliance checking

8. PRODUCT RELEASE
- Final product
- Certification
- Market launch

EMBEDDED PRODUCT DEVELOPMENT CYCLE

Leads to

PRODUCT QUALITY
- Reliable & stable product
- Better performance
- Fewer defects

TIME-TO-MARKET
- Efficient development
- Early testing & feedback
- Faster release

REGULATORY COMPLIANCE
- Safety standards
- Legal requirements
- Certification (e.g., CE, ISO)